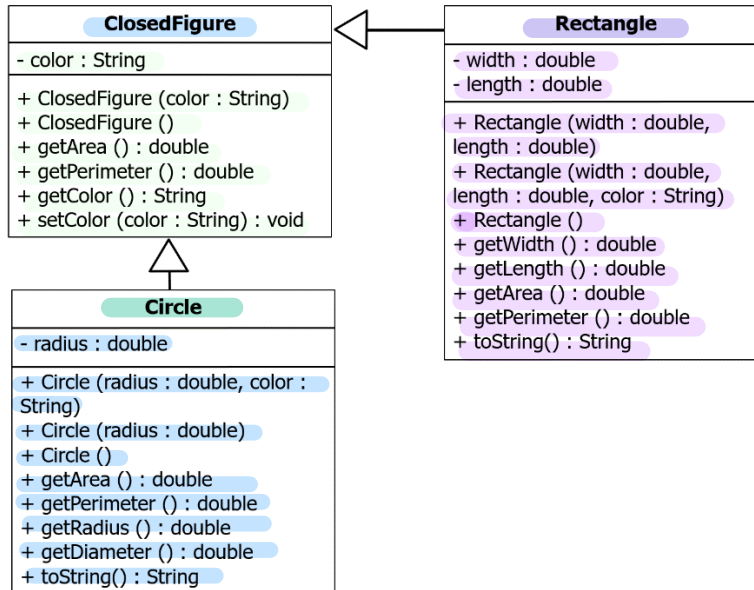




Quiz #09 Inheritance (20/04/2023)

1. จงเขียนโปรแกรมเพื่อทำการ Inheritance ข้อมูลคลาสจาก UML คลาสที่กำหนดมาให้พร้อมทั้งเขียนโปรแกรมสำหรับใช้ในการทดสอบ



โดยโปรแกรมทดสอบจะทำการสร้างอ็อบเจกต์และป้อนข้อมูลตามข้อความตัวอย่างนี้

```

ClosedFigure c1 = new Circle(2.0);   ClosedFigure c2 = new Circle();
ClosedFigure c3 = new Circle(3.0, "red");

ClosedFigure r3 = new Rectangle();   ClosedFigure r1 = new Rectangle(3.0, 3.3, "blue");
ClosedFigure r2 = new Rectangle(2.0, 2.2);

System.out.println(r3); System.out.println(r1); System.out.println(r2);
System.out.println(c2); System.out.println(c3); System.out.println(c1);
    
```

ผลลัพธ์ที่จะแสดงบนหน้าจอ

```

(width, length, area, perimeter, color) = (1.0, 1.0, 1.0, 4.0, White)
(width, length, area, perimeter, color) = (3.0, 3.3, 9.899, 12.6, blue)
(width, length, area, perimeter, color) = (2.0, 2.2, 4.4, 8.4, White)
(radius, diameter, area, perimeter, color) = (1.0, 2.0, 3.1416, 6.283, White)
(radius, diameter, area, perimeter, color) = (3.0, 6.0, 28.274, 18.849, red)
(radius, diameter, area, perimeter, color) = (2.0, 4.0, 12.566, 12.566, White)
    
```

หน้า
main

```
D:\2\cpe121\8\test8_1.java
File Edit Tools Project Language Level Help
New Open Save Close Cut Copy Paste Undo Redo Find Compile

circle.java
CloseFigure.java
Rectangle.java
test8_1.java

1 class test8_1{
2     public static void main(String[] args){
3         CloseFigure c1 = new circle(2.0);
4         CloseFigure c2 = new circle();
5         CloseFigure c3 = new circle(3.0,"Red");
6         CloseFigure r3 = new Rectangle ();
7         CloseFigure r1 = new Rectangle (3.0,3.3,"blue");
8         CloseFigure r2 = new Rectangle (2.0,2.2);
9         System.out.println(r3);
10        System.out.println(r1);
11        System.out.println(r2);
12        System.out.println(c2);
13        System.out.println(c3);
14        System.out.println(c1);
15    }
16 }
17 }
```

```
D:\2\cpe121\8\CloseFigure.java
File Edit Tools Project Language Level Help
New Open Save Close Cut Copy Paste Undo Redo Find Compile Reset Run Test Javadoc Code Coverage

circle.java
CloseFigure.java
Rectangle.java
test8_1.java

1 public class CloseFigure{
2     private String color;
3     public CloseFigure (String color){
4         this.color = color;
5     }
6     public CloseFigure (){
7         this.color = "white";
8     }
9     public double getArea(){
10        return 0.0;
11    }
12    public double getPerimeter(){
13        return 0.0;
14    }
15    public String getColor(){
16        return color;
17    }
18    public void setColor (String color){
19        this.color = color;
20    }
21 }

// สีเริ่มต้นสีขาว
// ได้ เลขอะไรก็ได้
// ได้ เลขอะไรก็ได้
// set สีใหม่
```

```
D:\2\cpe121\8\Rectangle.java
File Edit Tools Project Language Level Help
New Open Save Close Cut Copy Paste Undo Redo Find Compile Reset Run Test Javadoc Code Coverage

circle.java
CloseFigure.java
Rectangle.java
test8_1.java

1 public class Rectangle extends CloseFigure {
2     private double width;
3     private double length;
4     public Rectangle (double width,double length){
5         this.width= width;
6         this.length= length;
7     }
8     public Rectangle (double width,double length, String color){
9         super(color);
10        this.width = width;
11        this.length = length;
12    }
13    public Rectangle(){
14        this.width=1;
15        this.length=1;
16    }
17    public double getWidth(){
18        return width;
19    }
20    public double getLength(){
21        return length;
22    }
23    public double getArea(){
24        return width*length;
25    }
26    public double getPerimeter(){
27        return 2*(width+length);
28    }
29    public String toString(){
30        return "(width,length,area,perimeter,color) = (" +getWidth()+","+getLength()+","+getArea()+","+getPerimeter()+","+getColor()+")";
31    }
}
```

